Business Process Evolution

An Assignment on the Evolution of Business Processes in Leading Companies

Submitted by: Gule Narjis

Registration No: SP23-BSE-023

Course: Business Process Engineering

Instructor: Mukhtiar Zamin

COMSATS University Islamabad

Abbottabad Campus

Date: 21-10-2026

1. Introduction

A business process is a set of related activities that takes one or more inputs and turns them into an output that has value for a customer. Ordering a book online, admitting a patient to a hospital or processing an insurance claim are all business processes. The customer rarely sees the steps in between, but those steps decide whether the output arrives quickly, cheaply and without errors. As Dumas and his co-authors put it in Fundamentals of Business Process Management, every organisation, whether it realises it or not, runs on its processes.

Processes do not stay still, because the world around them does not. Three forces push them to change. The first is technology: the internet, barcode scanning, warehouse robots and machine learning have each made steps possible that were impractical twenty years ago. The second is the customer. People who once accepted a week of waiting for a parcel now expect it the same day, and viewers who once drove to a video shop now expect a film to start in seconds. The third force is competition and scale. A method that works for a hundred orders a day usually breaks at a hundred thousand, and a rival with a faster process takes the customers of the one who stands still.

To study this evolution properly we need a common language for drawing processes. Business Process Model and Notation (BPMN), maintained by the Object Management Group (OMG), is that language. A BPMN diagram uses a small set of standard symbols, so a diagram drawn by one person can be read by analysts, managers and software engineers alike. The table below summarises the elements used in this assignment.

Element	**What it looks like**	**Meaning**
Start event	Thin circle	Where the process begins; it is triggered by something, for example a customer placing an order.
End event	Thick circle	Where one path of the process finishes.
Task / activity	Rounded rectangle	A unit of work performed by a person, a team or a system.
Exclusive gateway	Diamond with an X (or unmarked)	A decision point where exactly one outgoing path is taken, based on a condition.
Sequence flow	Solid arrow	Shows the order in which activities happen.
Message flow	Dashed arrow	Shows communication between two separate participants, for example a company and a postal service.
Pool and lane	Large rectangle divided into rows	A pool represents one participant (a company or a customer); lanes divide it into roles or departments.
Data object	Document symbol	Information or a physical item that is created or used by a task, such as a pick list or an order record.
Text annotation	Bracketed note attached to the diagram	Extra explanation that does not affect the flow.

This assignment first profiles five large companies, then examines two processes in depth, Amazon's order fulfilment and Netflix's content delivery, comparing each process's first version with its latest version through BPMN diagrams.

2. Task 1: Top Five Companies

The five companies selected are listed below. They were chosen because each one is a leader in its industry and because each owes much of that position to how well it runs its processes.

Company	Headquarters	Type	Industry	Why it is big
Amazon	Seattle, United States	Products and services	E-commerce and cloud computing	Net sales of about US$638 billion in 2024; one of the world's largest online retailers and the leading cloud provider through AWS.
Toyota	Toyota City, Japan	Products	Automotive	Sold about 10.8 million vehicles in 2024, more than any other carmaker, for the fifth year in a row.
Netflix	Los Gatos, United States	Services	Media and entertainment (streaming)	Ended 2024 with about 301.6 million paid memberships in more than 190 countries; the world's largest subscription streaming service.
Apple	Cupertino, United States	Products and services	Consumer electronics and digital services	Net sales of about US$391 billion in fiscal 2024; the iPhone alone accounts for roughly half of its revenue.
McDonald's	Chicago, United States	Services	Fast-food restaurants	More than 43,000 restaurants in over 100 countries at the end of 2024, around 95 per cent of them franchised.

Amazon

Amazon was founded by Jeff Bezos in 1994 and began as an online bookshop run from a garage in Seattle. It now sells almost everything, runs the Prime subscription service and, through Amazon Web Services, rents computing power to much of the internet. Net sales reached roughly US$638 billion in 2024 (Amazon, 2025), and the company employs around 1.5 million people. What makes Amazon interesting for this assignment is that its size rests on logistics: hundreds of millions of orders must be picked, packed and delivered accurately, and the process behind that has been rebuilt several times.

Toyota

Toyota Motor Corporation, founded in 1937 and based in Toyota City, Japan, is the world's largest carmaker by volume. Including its Daihatsu and Hino subsidiaries, the group sold approximately 10.8 million vehicles in 2024, keeping the top position for the fifth consecutive year (Toyota Motor Corporation, 2025). Toyota is also famous inside management studies for the Toyota Production System, a set of ideas about waste reduction and just-in-time production that changed how factories everywhere are organised. Few companies show better that a well-designed process can itself be a competitive advantage.

Netflix

Netflix was founded in 1997 in Scotts Valley, California, by Reed Hastings and Marc Randolph, and it started life posting DVDs to subscribers. It closed 2024 with about 301.6 million paid memberships and revenue of roughly US$39 billion (Netflix, 2025). Netflix is a rare example of a company that deliberately replaced its own successful process, DVD rental by mail, with a completely different one, streaming, before competitors forced it to. That makes its content delivery process ideal for studying evolution.

Apple

Apple, founded in 1976 and headquartered in Cupertino, California, designs the iPhone, iPad and Mac and sells a growing range of services such as the App Store and iCloud. Its net sales were approximately US$391 billion in fiscal 2024, with the iPhone contributing about half (Apple Inc., 2024). Apple is known for keeping tight control over product design and for a supply chain that can build and ship tens of millions of devices around a single launch date.

McDonald's

McDonald's began as a small restaurant in San Bernardino, California, in 1940, and grew after Ray Kroc began franchising it in 1955. By the end of 2024 it had more than 43,000 restaurants in over 100 countries, and about 95 per cent of them were run by independent franchise owners rather than by the company itself (McDonald's Corporation, 2025). Its original Speedee Service System, an assembly-line method of preparing food, is one of the earliest famous examples of a business process being deliberately redesigned, and the company's standardised procedures are what let a burger taste the same in Chicago and in Karachi.

3. Task 2: Evolution of Two Business Processes

This section studies one process from two of the companies above. For each process, the story of its evolution is told first; then the earliest version and the latest version are drawn as BPMN diagrams and compared, with the information artefacts and the resources (lanes) of the process listed in tables.

3.1 Amazon: The Order Fulfilment Process

Order fulfilment is everything that happens between a customer clicking "buy" and the parcel arriving at the door. It is Amazon's core operational process, and it has passed through four broad stages.

Stage 1: The manual years (1995-1999). Amazon's first warehouse was a single facility near Seattle. Orders were printed on paper, staff walked the aisles with the printed pick list, collected the books by hand, packed them at a bench and handed the parcels to an outside courier. The process was simple and worked at bookshop volumes, but it was slow, and picking errors were common because nothing was checked by a machine.

Stage 2: The fulfilment centre network and Prime (2000s). As volumes grew, Amazon built large fulfilment centres close to major cities, so that stock sat nearer to customers. Barcode scanning and warehouse software replaced the paper list. The launch of Prime in 2005, which promised fast, free delivery for a yearly fee, put real pressure on this process: the promise was only affordable if picking and dispatch became much quicker and the network much denser.

Stage 3: Robotics after the Kiva acquisition (2012 onwards). In 2012 Amazon bought Kiva Systems, a warehouse robotics company, for about US$775 million. Kiva's orange robots reversed the old logic of fulfilment: instead of people walking to shelves, mobile robots carry whole shelving pods to a stationary picker. Walking time, the largest waste in the old process, largely disappeared, and reported click-to-ship times fell from over an hour to roughly fifteen minutes.

Stage 4: The process today. Fulfilment is now orchestrated by software end to end. Machine learning forecasts demand and moves stock into the fulfilment centre nearest to the customers most likely to order it, often before any order exists. When an order arrives, the system allocates it, robots fetch the goods, items are scanned at every step, and sorting, route planning and live tracking are automated. Same-day delivery is routine in many cities.

Version 1: how fulfilment worked around 1997

Figure 1 shows the original process. Three participants are involved: the customer, the Amazon warehouse and an external delivery partner. The only information artefact is paper. Notice how much of the diagram is people physically moving objects, and that the customer hears nothing between ordering and delivery.

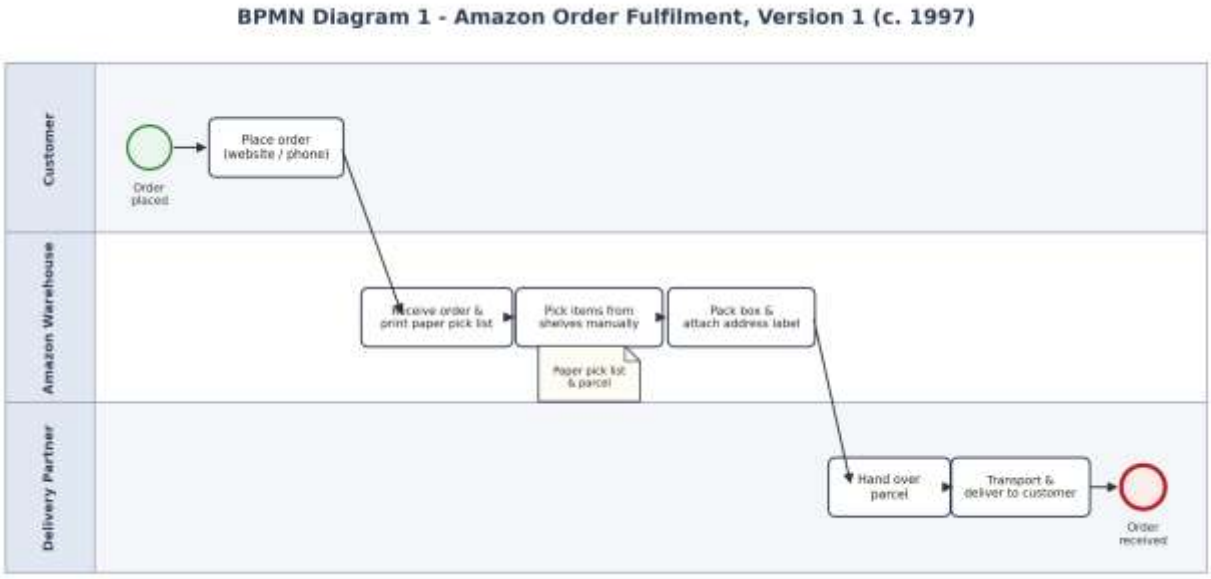


Figure 1: BPMN diagram of Amazon's order fulfilment process, Version 1 (c. 1997).

Latest version: fulfilment today

Figure 2 shows the same process now. Two features stand out. First, the Amazon platform lane contains a step that is not triggered by the order at all: demand forecasting runs continuously and pre-positions stock. Second, the decision gateway, "Item in stock at nearest fulfilment centre?", routes the order either to robotic picking locally or to a regional centre. The customer's side of the process has shrunk to a single click, while the company's side has become almost entirely system-driven.

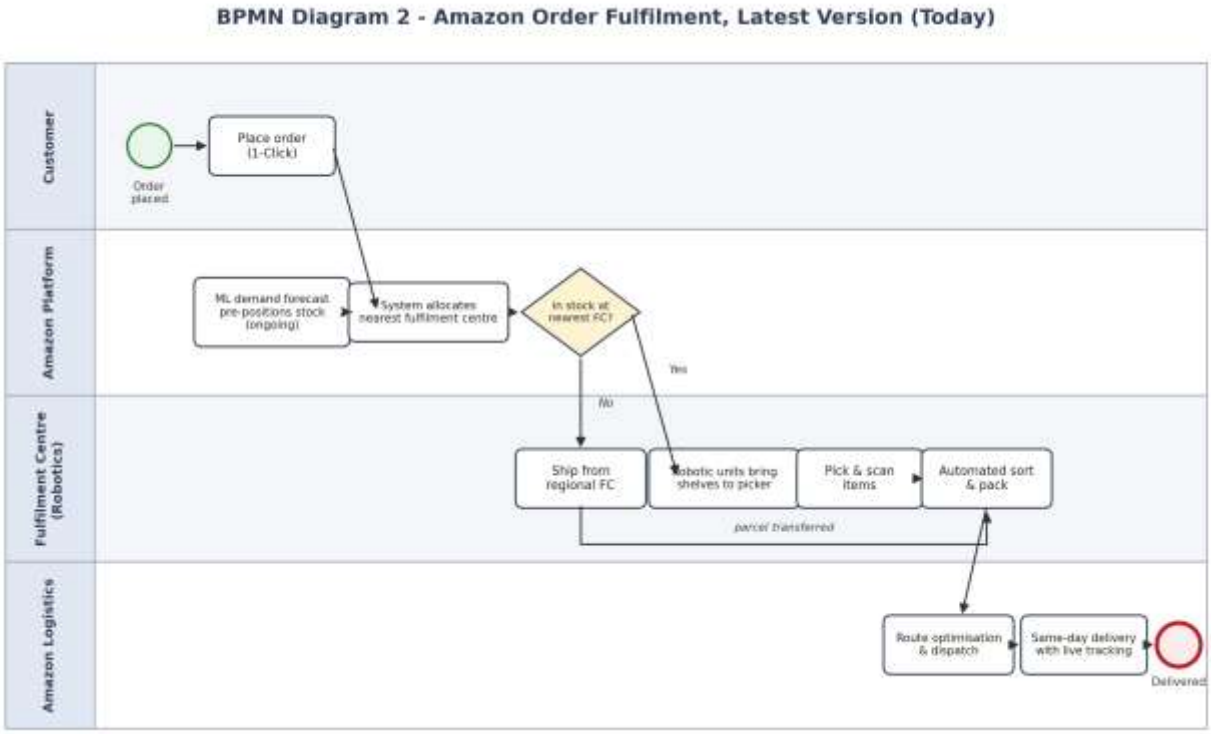


Figure 2: BPMN diagram of Amazon's order fulfilment process, latest version.

Comparison of the two versions

Aspect	**Version 1 (c. 1997)**	**Latest version**
Order capture	Order printed out on paper in the warehouse	Order enters the platform instantly; allocation is automatic
Picking	Staff walk the shelves holding a paper pick list	Robots bring shelving pods to a stationary picker
Record keeping	Paper lists and labels; errors found late	Every item scanned; inventory is a live database
Stock placement	One or a few warehouses holding most stock	Stock distributed across many centres, positioned by demand forecasts
Delivery speed	Several days, by an external courier	Next-day as standard; same-day in many cities, with live tracking
Role of people	People perform nearly every step	People supervise, pack and handle exceptions; systems and robots do the routine work
Scalability	More orders simply meant more walking and more paper	Scales through software, robotics and adding fulfilment centres

Information artefacts

Artefact	Produced by	Used by
Paper pick list (Version 1)	Warehouse office, from the printed order	Picker, to find the items on the shelves
Parcel with address label	Packing bench	Delivery partner, to route and deliver the order
Digital order record	Amazon platform, when the customer checks out	Fulfilment centre and Amazon Logistics, to allocate, pick and deliver
Live inventory record	Scanning tasks in the fulfilment centre	Allocation system and the website, to show availability and promise a delivery date
Demand forecast	Machine learning models on the platform	Inventory planners and the allocation system, to position stock before orders arrive
Tracking notification	Amazon Logistics	Customer, to follow the parcel in real time

Resources (lanes)

Resource / lane	Role in the process
Customer	Places the order and receives the parcel.
Amazon platform	Takes the order, forecasts demand, allocates stock and coordinates every later step.
Fulfilment centre (robotics)	Stores the goods; robotic units move shelves, associates pick, scan, sort and pack.
Amazon Logistics	Plans the route, dispatches the van and delivers, providing live tracking.
Delivery partner (Version 1 only)	An external courier that collected packed parcels and delivered them.

3.2 Netflix: The Content Delivery Process

Content delivery is the process of getting a film or series from Netflix to the viewer's screen. No process in this assignment changed more completely: in the first version the "content" was a physical disc travelling through the post; in the latest version nothing physical moves at all.

Stage 1: DVD by mail (1997-2007). Netflix launched in 1997 as a DVD rental service; the website opened in 1998 and a subscription model, unlimited rentals with no late fees, followed in 1999. Subscribers built a queue of titles on the website. Distribution centres posted the first available DVD in a red envelope; when it came back it was inspected and the next title in the queue was sent out. The whole process was hostage to the postal service: even with regional centres keeping turnaround near one day, watching a film meant planning days ahead.

Stage 2: Streaming begins (2007). In January 2007 Netflix added streaming, at first as a free extra for DVD subscribers and with only about a thousand titles. Broadband speeds limited the picture quality, but the delivery process lost its slowest steps in one move: no stock to wait for, no envelope, no return trip.

Stage 3: Open Connect (from about 2012). To keep millions of simultaneous streams smooth, Netflix built its own content delivery network, Open Connect, placing its servers inside or beside internet providers' networks so that popular titles are cached close to viewers. Adaptive streaming adjusts the picture quality moment by moment to suit each connection. Delivery was no longer something Netflix rented from the general internet; it became a process the company designed and controlled itself.

Stage 4: The process today. Delivery is instant, global and personal. The moment a subscriber opens the app, machine learning models decide which titles to show and in what order. Viewing data flows back continuously and improves those recommendations. The DVD process that started the company was finally shut down in 2023; the physical process and the company that runs the digital one share a name and a catalogue, and not much else.

Version 1: the DVD-by-mail process

Figure 3 shows the DVD-era process. The gateway "DVD available?" is the heart of it: the most wanted discs were often out with other subscribers, so a queue, and waiting, were built into the process design. The postal service is a full participant with its own lane, because the process cannot move faster than the post.

BPMN Diagram 3 - Netflix Content Delivery, DVD-by-Mail Era (c. 2005)

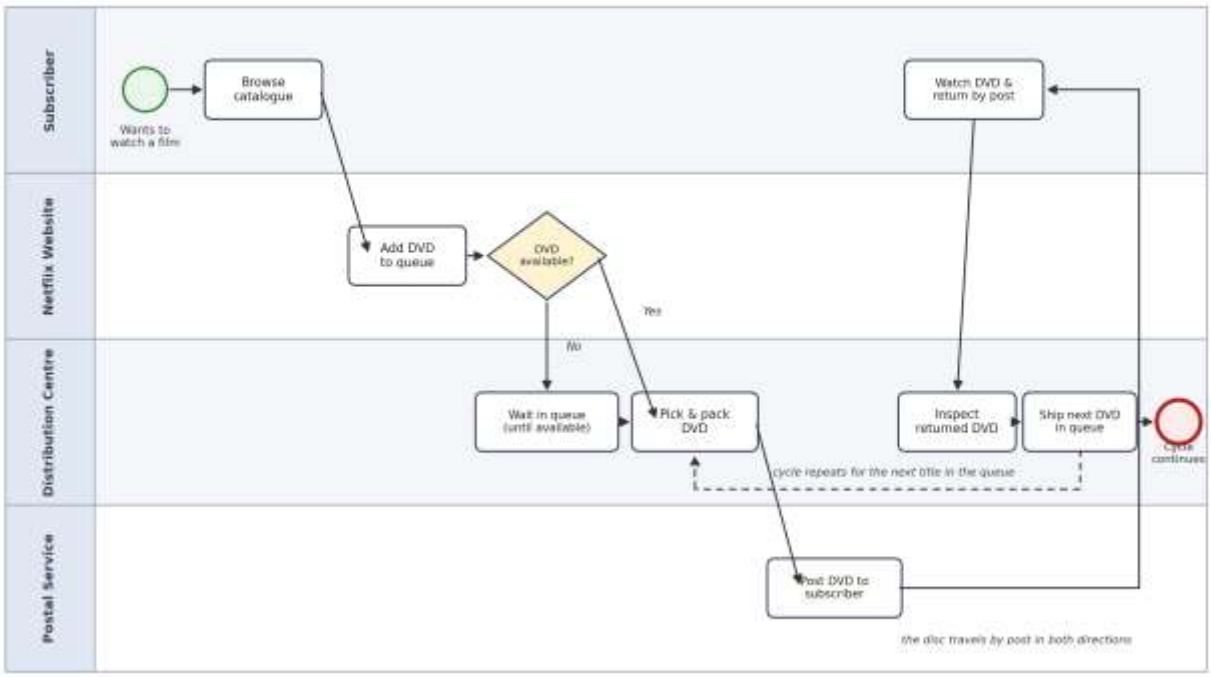


Figure 3: BPMN diagram of Netflix's content delivery process, DVD-by-mail version (c. 2005).

Latest version: the streaming process

Figure 4 shows content delivery today. The queue, the distribution centre and the postal service have all vanished. In their place are two things the DVD process never had: a recommendation step before the customer even chooses, and a feedback loop afterwards, in which watching itself produces data that improves the next session. The only physical infrastructure left, the Open Connect servers, sits quietly in the background of the diagram.

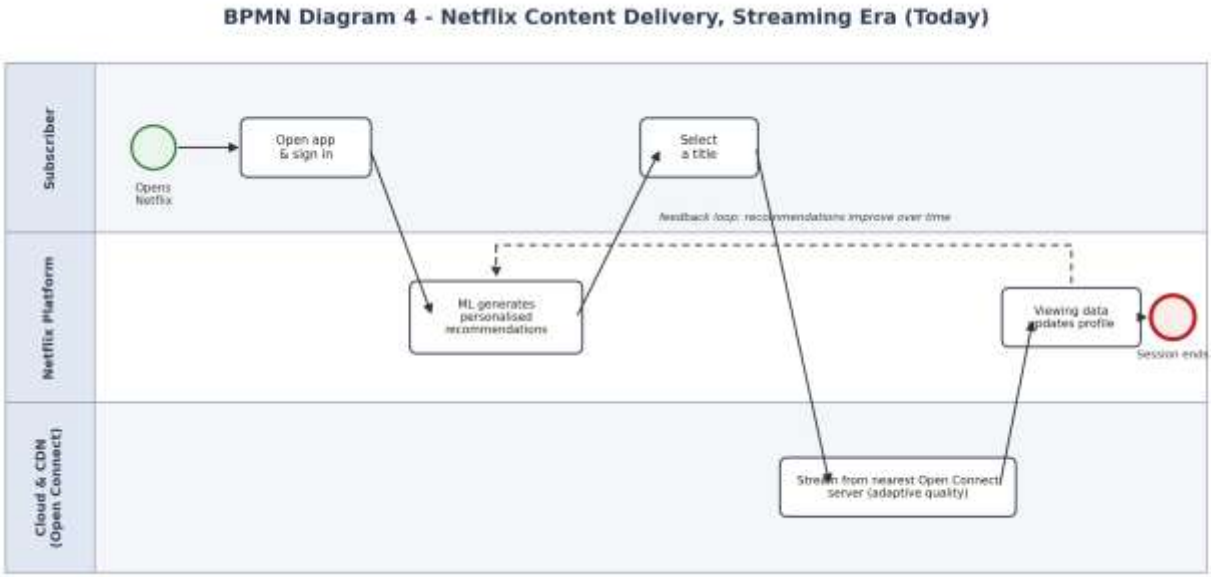


Figure 4: BPMN diagram of Netflix's content delivery process, streaming version (today).

Comparison of the two versions

Aspect	DVD version (c. 2005)	Streaming version (today)
What is delivered	A physical DVD, owned by Netflix and lent out	A digital stream; nothing physical changes hands
Time from choosing to watching	One to several days, depending on the post	Seconds
Availability	Limited by the number of discs owned; popular titles meant a wait in the queue	Effectively unlimited; the same title can stream to millions at once
Middle participants	Distribution centre staff and the postal service	Open Connect servers inside ISP networks
Personalisation	Same catalogue pages for everyone; a basic recommendation list	The catalogue itself is re-ordered for each viewer by machine learning
Feedback to the	Ratings and return envelopes	Continuous viewing data: what was

company		watched, paused, abandoned or rewatched
Main bottleneck	Postal speed and disc availability	The viewer's internet connection and content licensing

Information artefacts

Artefact	Produced by	Used by
Subscriber queue (DVD version)	Subscriber, on the Netflix website	Distribution centre, to decide which DVD to post next
DVD in its mailer envelope	Distribution centre	Postal service and subscriber; the envelope doubles as the return packaging
Inspection record (DVD version)	Distribution centre, when a disc returns	Inventory system, to return the disc to stock or retire it
Viewer profile and viewing history	Netflix platform, from sign-ins and viewing	Recommendation models, to personalise the catalogue
Cached content on Open Connect appliances	Netflix platform, copied to servers near viewers	Subscriber's device, which streams from the nearest copy
Recommendation list	Machine learning models on the platform	Subscriber, as the home screen they browse

Resources (lanes)

Resource / lane	Role in the process
Subscriber	Builds the queue or selects a title, watches the content, and (in the DVD version) returns the disc.
Netflix website / platform	Holds the catalogue and the queue; in the streaming version also runs the recommendation models and the viewer profile.
Distribution centre (DVD version)	Stores the discs, picks and packs them, and inspects returns.
Postal service (DVD version)	Carries the disc in both directions; its speed sets the speed of the whole process.
Cloud and CDN - Open Connect (streaming version)	Stores cached copies of titles close to viewers and streams them at a quality matched to each connection.

4. Task 3: Submission of the Solution

In line with the assignment instructions, this solution has been saved as a PDF and uploaded to cloud storage (Google Drive). The document can be opened through the following link: [PDF

Link]. The link is set so that anyone with it can view the file, and the same PDF is the version submitted for marking.

5. Task 4: Peer Review and Comparison

After the solutions were shared in class, I compared my work with that of my classmates. The comparison below uses five criteria: the companies selected, the depth of the process analysis, the quality of the BPMN diagrams, the treatment of artefacts and resources, and the insight drawn about how the processes evolved.

Note: the names and a few details in this section are placeholders. They will be updated with the actual names and observations from the class review session before final submission.

Criterion	**My solution**	**Classmates' solutions (general)**	**Outcome / comment**
Company selection	Five global companies; Amazon and Netflix studied in depth	One classmate selected banking processes; another chose local retail businesses	Global companies offer well-documented processes, but the local examples showed processes I could not have found in books.
Process depth	Two processes, each in two versions with four stages of evolution	Some classmates covered more processes but only at a surface level	Depth beat breadth: comparing two versions of the same process showed change more clearly.
BPMN quality	Pools and lanes for every participant; gateways used for real decisions	A few diagrams used gateways as decoration rather than for decisions, and some had no lanes	Lanes matter. Without them it is hard to see who is responsible for a step.
Artefacts and resources	Separate tables for artefacts and for lanes/resources in both processes	Most classmates mentioned resources inside the text but did not tabulate artefacts	Tabulating artefacts exposed steps I had missed, such as the DVD inspection record.
Evolution insight	Focused on what triggered each change: a promise (Prime), an acquisition (Kiva), a new channel (streaming)	Several solutions listed changes without explaining what caused them	Asking "what forced this change?" turned out to be the most useful question in the whole assignment.

[Classmate Name] selected a banking process, the opening of a new account, and traced how it moved from paper forms and in-branch verification to online applications with digital identity checks. That solution was stronger than mine on one point: it showed how regulation shapes a process, because every removed step had to be replaced by a compliance check somewhere else.

My two processes are driven by speed and convenience; the banking example was a useful reminder that some processes evolve under legal pressure, not customer pressure.

Another classmate, [Classmate Name], chose a local retail business and its stock-ordering process. The diagrams were simpler than mine and the process had barely changed in twenty years, which at first looked like a weakness. In the discussion it became the most interesting case in the room: it showed that process evolution is not automatic. Without competition, capital or technical skills, a process can stay manual for decades, and the business survives because its customers value other things. That contrast sharpened my own conclusions about Amazon and Netflix, which changed because they had both the reason and the resources to change.

Outcomes of the comparison:

- My BPMN diagrams held up well, mainly because every participant had a lane and every gateway answered a real question.
- I underplayed regulation and cost as drivers of change; the banking solution covered both better, and I have kept that in mind for the conclusion below.
- Classmates who used local businesses had weaker sources but stronger original observations; published case studies are detailed, yet they describe what a company says about itself.
- Across all the solutions reviewed, the same pattern appeared: the processes that changed most were the ones closest to the customer.

6. Conclusion

Comparing the first and latest versions of these two processes shows that business process evolution is rarely about one big invention. Amazon still picks, packs and delivers, just as it did in 1997; Netflix still delivers films chosen from a catalogue, just as it did in 1999. What changed is who, or what, does the work: paper gave way to live data, walking gave way to robots, the postal service gave way to a content delivery network, and judgement calls that people once made are now made continuously by software. In both cases the trigger was a promise made to the customer, faster delivery on the one hand and instant watching on the other, and the process had to be rebuilt to keep that promise at scale. The lesson I take from this assignment is that a process is never really finished; it is only ever the current answer to what customers expect and what technology allows.

7. References

Amazon.com, Inc. (2025). Annual Report 2024 (Form 10-K). U.S. Securities and Exchange Commission.

Apple Inc. (2024). Form 10-K for the fiscal year ended September 28, 2024. U.S. Securities and Exchange Commission.

Dumas, M., La Rosa, M., Mendling, J. and Reijers, H. A. (2018). Fundamentals of Business Process Management. 2nd edn. Berlin: Springer.

McDonald's Corporation (2025). Annual Report 2024 (Form 10-K). U.S. Securities and Exchange Commission.

Netflix, Inc. (2025). Q4 2024 Shareholder Letter and Annual Report 2024 (Form 10-K).

Netflix Technology Blog (2016). 'Open Connect: Netflix's Content Delivery Network'. Available at: netflixtechblog.com.

Object Management Group (2013). Business Process Model and Notation (BPMN), Version 2.0.2. OMG Document formal/2013-12-09.

Toyota Motor Corporation (2025). 'Results for December 2024: Global Sales, Production and Export Results for 2024'. Press release, Toyota City.

Weske, M. (2019). Business Process Management: Concepts, Languages, Architectures. 3rd edn. Berlin: Springer.